Point-Process Models for Order-Book Simulation

Neural Hawkes · s2p2 · State-Dependent Parallel Neural Hawkes · Compound Multivariate Hawkes

What a good order-book simulator needs (the axes used here):

0. STABILITY & CALIBRATION   free-rollout stays bounded at the right rate (no explode / no Poisson collapse).
1. STYLIZED FACTS   clustering (Fano), heavy tails, volatility long-memory (|r|-ACF), efficiency.
2. MECHANISTIC FIDELITY   genuine self/cross-excitation; book-consistent events; multivariate structure.
3. ACTION-CONDITIONALITY   reacts to the agent's own orders with realistic market impact (world-model use).

Key distinction throughout: a point-process model can be a strong next-event PREDICTOR yet a poor SIMULATOR (matching one-step targets does not imply a faithful long-horizon rollout). The branching ratio ρ governs stationarity: for a LINEAR Hawkes $\rho < 1$ is exact (iff); for a NONLINEAR link it is only sufficient.

Unified template: $\lambda_k(t) = f\left(\mu_k + \sum_{m,j} A_{kj}^m S_j^m(t)\,;\ \text{state}\right), \quad S_j^m(t) = \sum_{t_i < t,\ c_i = j} e^{-\beta_m (t - t_i)}$

Models differ in the link f, the state/memory, the number of timescales, and how they are fit.

Compiled from the TFOW / VolumeSetMTPP investigation on Gemini ETH event-driven data, 2026-06.

1. Functionality comparison matrix

Order-book simulation: formula & functionality comparison

Dimension	Neural Hawkes	s2p2	State-Dep Parallel NH	Compound MV Hawkes
Intensity form	f(w^Th(t)), h = CT-LSTM hidden, per-event gated	softplus(w^Tu^(L)), u = stacked latent SSM	f(h(t), s_book(t)): history + LOB state	mu_k + sum_i alpha e^{-beta(t-ti)}
Link f	softplus (phi->1)	softplus, after LayerNorm	softplus / exp	identity (LINEAR)
Memory / state	continuous-time LSTM cell, learned decay	stacked diagonal linear SSM (latent), parallel scan	parallel neural-Hawkes hidden + book features	exp-decayed per-type event counts (explicit)
Multi-timescale	learned per-dim cell decay	delta-spectrum + stacking	per-dim decay	single beta (one scale)
Branching / stability certificate	none (implicit)	gauge-BROKEN (LayerNorm scale-invar.)	none / implicit	EXACT rho = sr(alpha/beta)
Simulation method	Ogata thinning	autoregressive (inv-CDF dt + mark)	thinning / autoreg, book replay	Ogata thinning
LOB book-state dependence	no	optional (lob_state head)	YES — core design	no
Order size / volume	no (type only)	log-normal head (opt.)	paper-dependent	log-normal (the 'compound' part)
Calibration / training	MLE, sequential	windowed SGD (cold-start)	parallel SGD	full-stream MLE (exact likelihood)
Long-memory |r|-ACF (F6/F8)	partial (LSTM)	partial (F6 lifted)	partial	none (F6~0, flat)
Sim Fano(1s) (real ~ 8)	(not run here)	41 (over-disperses)	(not run; cf. PCT-LSTM 7.9)	8.8 (OK)
Strength / acc	expressive baseline	0.319 — best PREDICTOR	book-conditioned sim	0.18 — certified STABLE simulator

2. Intensity formulas (1/2)

Order-book simulation: formula & functionality comparison

Neural Hawkes (Mei & Eisner, 2017)

$$\lambda_k(t) = \mathrm{softplus}\big(w_k^\top h(t)\big), \quad h(t) = o_i \odot (2\sigma(2c(t)) - 1)$$
$$c(t) = \bar{c}_i + (c_i - \bar{c}_i)\, e^{-\delta_i (t - t_i)} \quad (t_i < t \le t_{i+1})$$

- Continuous-time LSTM: gates updated at each event; cell c(t) decays toward cbar between events.
- Per-dimension learned decay delta_i gives a soft multi-timescale memory.
- Marks: p(k|t) = lambda_k(t)/sum lambda(t). Simulation: Ogata thinning on the CT intensity.
- Very expressive, but NO branching/stationarity certificate — free rollout can drift; not LOB-aware.

s2p2 — stacked latent linear Hawkes (state-space PP)

$$\lambda(t) = \mathrm{softplus}\big(w^\top u^{(L)}(t)\big), \quad \frac{dx^{(\ell)}}{dt} = -\Lambda^{(\ell)} x^{(\ell)}, \quad x^{(\ell)} \leftarrow x^{(\ell)} + E^{(\ell)} e_i$$
$$u^{(\ell)} = \mathrm{LayerNorm}(\mathrm{GELU}(C^{(\ell)} x^{(\ell)}) + \mathrm{skip})$$

- Diagonal LINEAR latent dynamics => parallel scan (fast training); delta-spectrum + stacking => longer memory.
- LayerNorm readout RATE-BOUNDS the intensity (implicit stability) BUT makes the branching ratio gauge-invariant => the weight-norm certificate is GAMEABLE (cannot be honestly measured).
- Best next-event predictor in our study (acc 0.319); but over-disperses in simulation (Fano 41 vs real 8).

2. Intensity formulas (2/2)

Order-book simulation: formula & functionality comparison

State-Dependent Parallel Neural Hawkes for LOB

$$\lambda_k(t) = f_k(\,g(h(t),\ s_{\mathrm{book}}(t)))$$
$$h(t) = \text{ parallel neural-Hawkes hidden state;} \quad s_{\mathrm{book}}(t) = \text{ (imbalance, depth, spread, queue)}$$

- Intensity conditioned on the CURRENT LOB STATE in addition to event history — the defining feature.
- 'Parallel' = parallelizable training (scan/attention) rather than sequential RNN => scalable.
- Explicitly targets LOB event-stream PREDICTION AND SIMULATION; state-dependence aids book consistency and the response of flow to book conditions (a step toward action-conditional / impact realism).
- (Structural form shown — exact parameterization per the cited paper; our closest run proxy is PCT-LSTM, a parallel continuous-time LSTM Hawkes: Fano 7.9 via dt-variability, acc 0.270.)

Compound Multivariate Hawkes (Jain et al., 2024)

$$\lambda_k(t) = \mu_k + \sum_{t_i < t} \alpha_{k,c_i} e^{-\beta(t - t_i)}, \qquad \rho = \mathrm{spec.rad}(\alpha/\beta)$$
$$\text{size} \mid \text{type } k \ \sim \ \text{LogNormal}(m_k, s_k)$$

- LINEAR multivariate exp-kernel Hawkes => EXACT branching ratio; mean rate Lambda=(I-alpha/beta)^-1mu closed-forn
- alpha>=0 (excitatory) => Ogata thinning + well-defined stationarity (rho<1). Fit by full-stream MLE.
- 'Compound': per-type log-normal order sizes — the volume mark.
- Nails clustering (Fano 8.8 ~ real) and is certifiably stable; BUT single beta => FLAT Fano-vs-scale, no |r|-ACF long memory (F6~0), and a shallow next-event predictor (acc 0.18).

3. Verdict for order-book SIMULATION

Order-book simulation: formula & functionality comparison

Stability / calibration (Layer 0)

- Compound MV Hawkes: exact rho<1 certificate + full-stream MLE => bounded at the right rate. STRONGEST.
- s2p2: LayerNorm gives implicit boundedness but no honest certificate; over-disperses.
- Neural Hawkes / State-Dep Parallel NH: expressive, no explicit certificate => rollout can drift.

Stylized facts (Layer 1)

- Clustering / Fano: any properly-fit Hawkes solves it at rho~0.6–0.8 (Compound 8.8 ~ real).
- Fano-vs-scale RISE: needs MULTIPLE timescales — single-beta Compound is flat; a delta-spectrum (s2p2 / multi-timescale Hawkes) bends it upward toward the real cross-scale clustering.
- Volatility long-memory (|r|-ACF, F6/F8): UNSOLVED by all four — needs directional/regime structure (sign-reinforcing excitation or a stochastic-volatility factor), not just count excitation.

Mechanism & book consistency (Layer 2)

- State-Dependent Parallel NH is the only one with book-state conditioning built in => best placed for book-consistent dynamics and the response of order flow to book conditions.
- Multivariate (cross-type) excitation: Compound MV Hawkes and s2p2 carry it; classic Neural Hawkes is generic.

Action-conditionality & market impact (Layer 3 — the world-model goal)

- None of the four, as published, model the agent's own orders + impact. State-dependence (SDPNH) is the closest hook; the natural next step is to condition the intensity on actions WITHOUT breaking the stability certificate (e.g. a bounded action-gate on a certified Hawkes backbone).

Bottom line

- Best PREDICTOR: s2p2 (deep, expressive). • Best certified SIMULATOR: Compound MV Hawkes.
- No single model gives prediction AND faithful simulation AND an honest certificate together.
- The synthesis pursued in this project: a certified linear multivariate-Hawkes BACKBONE (Compound MV) x a bounded neural GATE (s2p2 expressiveness, optionally book/action-state-dependent like SDPNH), trained full-stream so the rate is calibrated — stable AND expressive, with the certificate intact.